

Capstone Project Proposal

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Date: 5th June 2025

1. Problem Statement

We aim to build a regression model that predicts the final sale price of residential homes in Ames, Iowa, based on 79 features describing lot size, building quality, room counts, basement/fireplace/garage details, and more. This tool helps real estate agents generate accurate price estimates so sellers can set competitive listing prices and buyers can evaluate whether an asking price is fair.

2. Dataset

Name: House Prices — Advanced Regression Techniques
Source: Kaggle Competition
Link:
<https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data?select=train.csv>
Size: 1,460 rows × 81 columns (79 features + Id + SalePrice target)

Justification Against the 7 Criteria

#	Criterion	How the Dataset Meets It
1	Minimum size (1,000 rows, 8 features)	1,460 rows, 79 features — exceeds both thresholds
2	Public availability	Free on Kaggle with downloadable CSV
3	Real-world origin	Real residential sales records from Ames, Iowa (2006–2010)

#	Criterion	How the Dataset Meets It
4	Heterogeneous features	Mix of numerical (LotArea, YearBuilt, SalePrice), ordinal (OverallQual, KitchenAbvGr), categorical (MSZoning, Neighborhood, RoofStyle)
5	Imperfections	Missing values in ~20 columns (Alley, PoolQC, Fence, MiscFeature, etc.) + noisy values
6	Clear target/objective	Well-defined numerical target: SalePrice (supervised regression)
7	Ethics check	No names, IDs (except row IDs), GPS coordinates, or sensitive personal data

3. Preliminary Algorithms

Algorithm	Why Considered
Linear Regression	Simple baseline, interpretable coefficients
Ridge Regression	Handles multicollinearity (79 features likely correlated)
Random Forest Regressor	Non-linear, robust to outliers, feature importance
XGBoost Regressor	Usually top performer on tabular data

Evaluation Metrics: RMSE (primary), MAE, R^2 , RMSE in log space (as used by Kaggle)

4. Pipeline Sketch

Raw CSV → Load & merge → Data audit (missing, dtypes, stats)
→ Wrangling (impute/drop, encode, outlier removal)
→ EDA (8+ visualizations, insights)
→ Feature engineering (3+ new features)
→ Store in SQLite
→ Train/test split (80/20)
→ Train: Linear, Ridge, Random Forest, XGBoost
→ Hyperparameter tuning (GridSearchCV)
→ Evaluate & compare → Best model → Export .pkl
→ Model Card → Final report & presentation